



**PHYSICAL RESEARCH
LABORATORY**
PRL, Ahmedabad

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Atmospheric Visibility Estimation from Ground-Based Imagery

Why Estimate Atmospheric Visibility?

Atmospheric conditions change the contrast, clarity and visible range of an outdoor scene.

Atmospheric information

Haze, aerosols, humidity and fog scatter light, weakening landmark contrast and horizon clarity.

Monitoring relevance

Visibility provides an interpretable indicator for identifying and studying reduced-visibility atmospheric conditions.

Camera-based opportunity

A fixed-view camera enables inexpensive, repeatable and visually auditable observations that can be analysed using AI.

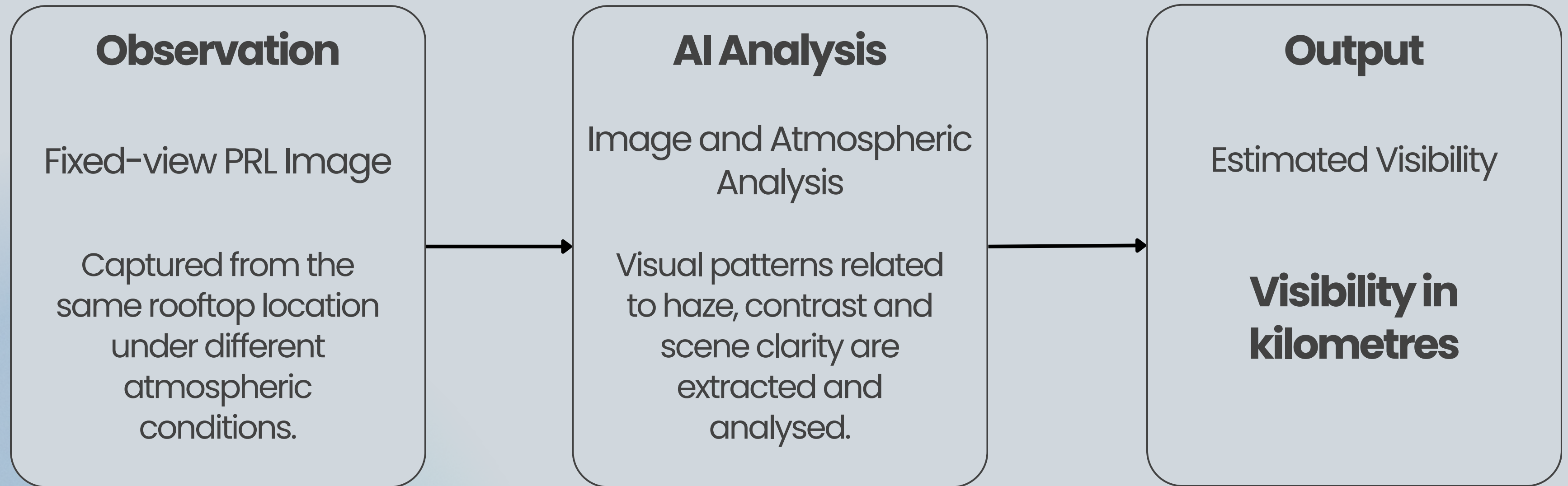


Fixed-view PRL scene used to observe atmospheric clarity and estimate visibility.

Can a model convert the visual appearance of a fixed PRL scene into a visibility estimate in kilometres?

Project Goal and Approach

Developing an AI-based research system to estimate atmospheric visibility from a fixed-view PRL image.



Research focus: Can visual scene information identify changes in atmospheric visibility, particularly during reduced-visibility conditions?

PRL Data Collection and Dataset



GoPro Hero 9 installed at a fixed rooftop observation point.

Same camera • Same viewpoint • Same reference scene

DATASET SUMMARY

Local images	127
Acquisition dates	23
Collection period	27 May–29 June 2026
Afternoon observations	2 PM–5 PM

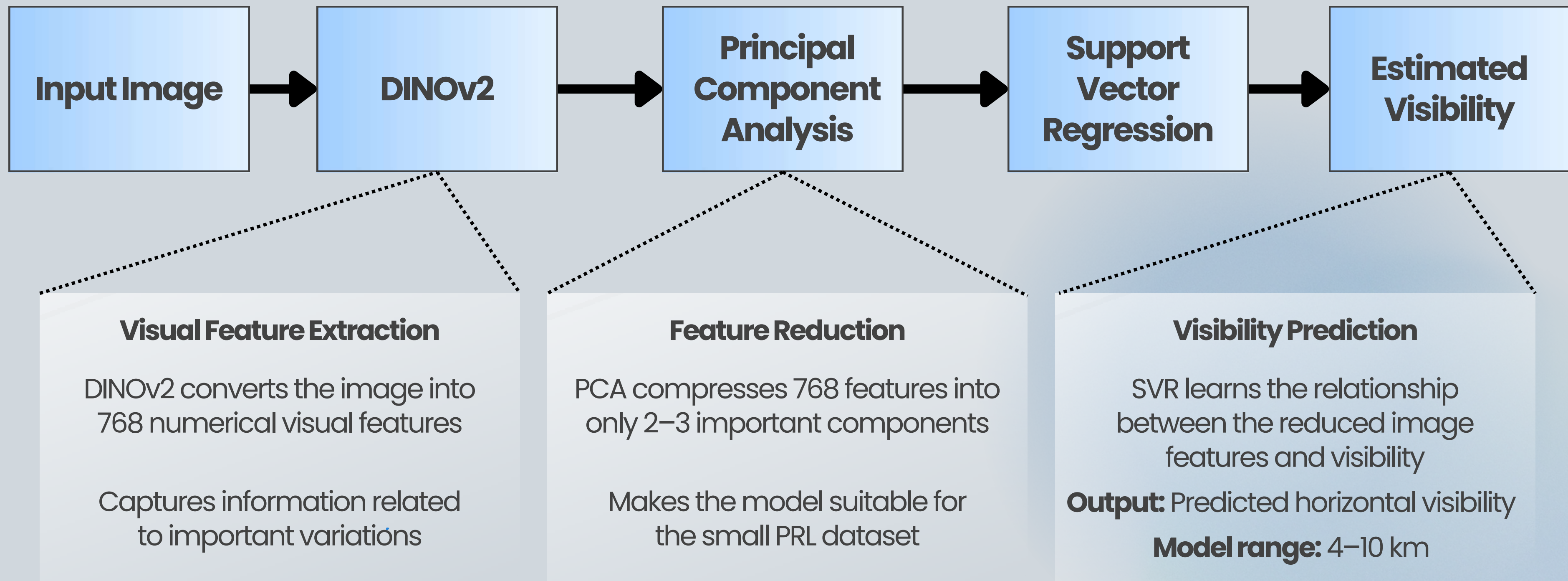
Image capture → Atmospheric observations → Aligned research dataset

INFORMATION RECORDED

Visibility	PRL, Ahmedabad
Temperature	
Relative humidity	23.036° N, 72.544° E
Solar intensity	
Date and time	Distance from Clock Tower – 265 Meters
Location	

Visibility Estimation Pipeline

The final system converts a fixed-view PRL image into a visibility estimate using frozen DINOv2 features, PCA and Support Vector Regression.



How the Final Model Was Selected

● SOURCE-DOMAIN PROOF OF CONCEPT

14,906 SkyFinder images

Images + Weather + Visibility labels

Model	MAE
Weather-only	2.071 km
DINOv2 image-only	2.005 km
Image + weather fusion	1.550 km

Best source result, $R^2 = 0.604$

● SELECTING THE PRL MODEL



Online dataset: Confirmed that visibility-related patterns can be learned

PRL dataset: Showed that the image-only PCA–SVR model generalized best locally

The Main Challenge: PRL Dataset Imbalance

The local dataset was carefully collected, but most observations represented the same clear-visibility condition.



[4 km image]



[10 km image]

Exact visibility distribution

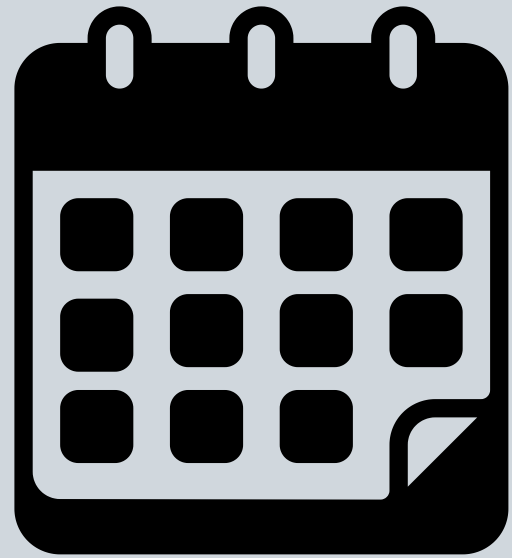
Visibility	Images	Percentage
4 km	1	0.8%
5 km	10	7.9%
9 km	11	8.7%
10 km	105	82.7%

105 of 127 images are labelled 10 km. Therefore, a model that always predicts 10 km already performs well overall. We must also check how well the model detects reduced visibility.

Only **23 independent observation dates** were available, not 127 completely independent conditions.

How We Tested the Model Fairly

The model was tested on complete acquisition dates that it had never seen during training.



23 observation dates
127 total images

[Images captured on the same date were treated as one group.]



22 TRAINING DATES
1 UNSEEN TEST DATE

DINOv2 features → PCA → SVR training
Predict visibility for the held-out date



Every date was used once as the unseen test date.

Final results combine predictions from all 23 held-out dates.

Why split by date?

Images from the same day have very similar lighting and atmospheric conditions. Keeping each date together prevents the model from seeing nearly identical conditions during both training and testing.

Final Results: Overall Model Performance

Performance on unseen acquisition dates
was assessed using MAE, RMSE and R².

What do these results mean?

The constant 10 km baseline achieved the lowest average error because 82.7% of the observations were labelled exactly 10 km. However, the balanced PCA–SVR achieved the lowest RMSE, indicating better control of larger prediction errors.

Reduced–visibility result:

For images below 10 km, the balanced PCA–SVR reduced MAE from 3.045 km to 2.491 km.

This shows useful sensitivity to hazy conditions, despite the constant baseline having the best overall MAE.

Overall Mean Absolute Error	
Average prediction error in kilometres – lower is better	
Model	MAE (km)
Constant 10 km baseline	0.528 km
MAE–selected PCA–SVR	0.651 km
Balanced PCA–SVR	0.731 km

Additional Performance Measures		
Model	RMSE (km) ↓	R ² ↑
Constant 10 km baseline	1.529	–0.135
MAE–selected PCA–SVR	1.454	–0.026
Balanced PCA–SVR	1.422	0.019

RMSE: Gives greater importance to large prediction errors
R²: Measures explained variation; higher is better

Limitations, Future Work & Conclusion

CURRENT LIMITATIONS

- 127 images across only 23 dates
- 82.7% of labels were 10 km
- Very few low-visibility examples
- Single camera and fixed viewpoint
- Short collection period
- No instrument-based validation

FUTURE WORK

- Extend collection across seasons and times
- Add multiple viewpoints and locations
- Validate using a visibility instrument
- Re-evaluate weather-image fusion
- Develop a real-time monitoring system
- Explore AOD estimation using AOD reference data

CONCLUSION

- Developed an end-to-end image-based visibility estimation pipeline
- Used date-wise evaluation to test unseen atmospheric conditions
- Learned model showed useful sensitivity below 10 km, but overall performance was limited by dataset imbalance

The project establishes a reproducible baseline for future camera-based atmospheric visibility monitoring at PRL.

Thank You

for your time and
attention